

AFFICIONADO COFFEE ROASTERS

Sales & Demand Analytics

Data Analytics Project Report

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Abstract

This project presents a data-driven analysis of sales transactions from Afficionado Coffee Roasters with the objective of identifying revenue patterns, customer purchasing behavior, product performance, store performance, and periods of peak demand. The analysis was performed using Python, Pandas, Matplotlib, Plotly, and Streamlit.

The dataset contains 149,116 transactions with a total quantity sold of 214,470 units and total recorded revenue of \$698,812.33. The analysis identified Hell's Kitchen as the highest-performing store, while Coffee was the leading product category by revenue. The analysis also identified Sustainably Grown Organic Lg as the top-performing product and 10:00 AM as the peak revenue hour.

An interactive Streamlit dashboard was developed to present the findings and allow users to explore transaction data, revenue performance, store-level performance, product categories, and hourly demand. The results provide useful insights that can support operational decisions related to staffing, inventory planning, product prioritization, and demand management.

EXECUTIVE SUMMARY

This project presents a data analytics and demand analysis study of Aficionado Coffee Roasters using transaction-level sales data. The objective of the project is to analyze historical coffee sales, identify important sales patterns, evaluate store and product performance, and determine the peak hours of customer demand.

The dataset contains approximately 149,116 transactions and includes information such as transaction ID, year, transaction time, transaction quantity, store location, product ID, unit price, product category, product type, and product details. The data was cleaned, transformed, and analyzed using Python and Pandas. Additional features such as hourly sales and revenue were created to support the analysis.

The analysis identified a total revenue of \$698,812.33 from 149,116 transactions, with a total quantity sold of 214,470 units. The average transaction value was approximately \$4.69. Among the stores, Hell's Kitchen generated the highest revenue, while Coffee was the highest-performing product category. The highest revenue-generating product was Sustainably Grown Organic Lg, and 10:00 AM was identified as the peak revenue hour.

An interactive dashboard was developed using Streamlit, Pandas, and Plotly to present the findings in a user-friendly format. The dashboard allows users to explore transaction performance, revenue trends, store performance, product categories, and hourly demand.

The results of this project can help coffee business managers understand customer purchasing behavior, identify high-performing products and locations, optimize staffing during peak hours, and make data-driven business decisions.

1. INTRODUCTION

1.1 Background

The coffee industry generates a large volume of transactional data through daily customer purchases. This data contains valuable information about customer purchasing patterns, product preferences, store performance, revenue generation, and demand fluctuations throughout the day. Analyzing such data can help businesses understand their operations and make informed decisions.

Aficionado Coffee Roasters provides a suitable dataset for studying these patterns because it contains detailed transaction-level information including transaction time, quantity sold, store location, product category, product type, product details, and unit price.

In this project, Python-based data analytics techniques are used to transform the raw transaction data into meaningful business insights. The analysis focuses particularly on revenue, transaction volume, product performance, store performance, and hourly demand.

1.2 Project Overview

The project focuses on developing a comprehensive sales and demand analytics system for Aficionado Coffee Roasters. The raw transaction dataset is first examined and prepared for analysis. Data cleaning and transformation techniques are then applied to ensure that the dataset is suitable for further analysis.

Python and Pandas are used for data processing and analysis, while visualization techniques are used to identify trends and patterns. Additional analytical variables, including revenue and hour of day, are generated from the original transaction data.

The final stage of the project involves developing an interactive web-based dashboard using Streamlit. The dashboard presents important business metrics and visualizations in an accessible format, allowing users to explore sales performance and demand patterns.

1.3 Need for the Study

Businesses operating in the food and beverage industry need to understand customer demand in order to optimize their resources and improve profitability. Simply collecting transaction data is not sufficient unless the information can be converted into useful insights.

This study is therefore conducted to identify important patterns within coffee sales data. Understanding which products generate the highest revenue, which stores perform better, and which hours experience the highest demand can support better operational and managerial decisions.

The study also demonstrates how data analytics can be applied to a real-world business dataset to support evidence-based decision-making.

1.4 Scope of the Project

The scope of this project includes the analysis of transaction-level coffee sales data and the development of an interactive analytical dashboard. The project covers data preparation, exploratory data analysis, revenue analysis, product analysis, store performance analysis, and hourly demand analysis.

The project also includes the creation of visualizations that communicate important patterns in the dataset. The final dashboard provides an interactive interface through which users can examine key business metrics and analytical results.

The project is primarily focused on descriptive and exploratory analytics. The findings are intended to help understand historical sales performance and demand patterns and can serve as a foundation for future predictive forecasting and advanced business analytics.

2. PROBLEM STATEMENT

2.1 Problem Statement

Coffee businesses generate a large amount of transaction data through daily customer purchases. However, raw transactional records alone do not provide meaningful information for business decision-making. Without systematic analysis, it can be difficult to identify high-performing products, profitable store locations, revenue patterns, and periods of high customer demand.

The problem addressed in this project is to transform raw coffee transaction data into meaningful analytical insights that can help understand sales performance and customer demand. The project focuses on examining transaction volume, revenue, product performance, store performance, and hourly demand patterns.

The analysis is intended to answer important business questions such as which store generates the highest revenue, which product category performs best, which products contribute significantly to revenue, and which hours experience the highest customer demand. These insights can help support better operational planning and data-driven decision-making.

2.2 Research Questions

1. What is the total revenue generated from the available transactions?
2. What is the total number of transactions and units sold?
3. Which store location generates the highest revenue?
4. Which product category contributes the most revenue?
5. Which individual product generates the highest revenue?
6. Which hours of the day experience the highest revenue and demand?
7. How can the identified sales patterns support business decision-making?
8. How can an interactive dashboard improve the presentation and exploration of the analysis?

2.3 Business Problem

The primary business challenge is to convert a large volume of historical sales transactions into information that can be easily understood and used by decision-makers. Management needs to understand where revenue is being generated, which products are performing well, and when customer demand is highest.

By analyzing these factors together, the business can identify opportunities for improving staffing, inventory planning, product promotion, and store-level performance. The project therefore applies data analytics techniques to provide a structured view of business performance.

3. PROJECT OBJECTIVES AND REQUIREMENTS

3.1 Primary Objective

The primary objective of this project is to analyze transaction-level sales data of Aficionado Coffee Roasters and develop an interactive analytics dashboard that provides meaningful insights into revenue, transaction volume, product performance, store performance, and hourly customer demand.

3.2 Specific Objectives

1. To understand and examine the structure of the coffee sales dataset.
2. To clean and preprocess the transaction data for analysis.
3. To calculate key performance indicators such as total revenue, total transactions, total quantity sold, and average transaction value.
4. To analyze revenue performance across different store locations.
5. To identify the highest-performing product categories and individual products.
6. To analyze hourly transaction and revenue patterns.
7. To identify the peak revenue-generating hour.
8. To create meaningful charts and visualizations for business analysis.
9. To develop an interactive dashboard using Streamlit.
10. To provide actionable business insights based on the analytical findings.

3.3 Functional Requirements

The developed analytics system should be capable of loading and processing the available coffee transaction data. It should allow the analysis of sales and revenue performance across different business dimensions such as store location, product category, product type, and time.

The system should also present important performance indicators and visualizations through an interactive dashboard. Users should be able to view and interpret key metrics and identify important sales and demand patterns.

The system should provide:

1. **Data Loading** – Ability to load the processed transaction dataset.
2. **Data Processing** – Ability to transform transaction and time-related fields.
3. **KPI Analysis** – Display of important business performance indicators.
4. **Revenue Analysis** – Analysis of revenue across stores, products, and categories.
5. **Hourly Analysis** – Identification of hourly sales and revenue patterns.
6. **Visualization** – Graphical representation of analytical results.
7. **Interactive Dashboard** – User-friendly presentation of results using Streamlit.

3.4 Non-Functional Requirements

The dashboard should be simple, interactive, readable, and easy to navigate. The analytical system should provide consistent results when the same dataset is processed. Visualizations should clearly communicate the underlying patterns without unnecessary complexity.

The system should also be maintainable so that additional transaction data can be incorporated in the future. The dashboard should provide an efficient interface for users who may not have technical knowledge of Python or data analysis.

4. METHODOLOGY

4.1 Research Methodology

The project follows a data analytics methodology to examine historical coffee sales transactions and identify meaningful patterns in revenue, transaction volume, product performance, store performance, and hourly demand.

The methodology begins with collecting and understanding the available transaction dataset. The raw data is then inspected for its structure, data types, missing values, and consistency. Data preprocessing and transformation techniques are applied to prepare the dataset for analysis.

Exploratory data analysis is subsequently performed to identify important trends and relationships within the data. Derived variables such as revenue and transaction hour are created to support more detailed analysis.

The processed data is then analyzed using descriptive statistical techniques and visualizations. Key performance indicators are calculated to summarize overall business performance. Finally, the analytical results are presented through an interactive Streamlit dashboard that allows users to explore the findings in a simple and user-friendly manner.

4.2 Data Collection and Dataset

The dataset used in this project contains transaction-level records of coffee sales from Afficionado Coffee Roasters. Each transaction represents a coffee shop sales event and contains information related to the transaction, store, product, quantity, and price.

The dataset contains 149,116 transaction records. The major attributes include transaction ID, year, transaction time, transaction quantity, store ID, store location, product ID, unit price, product category, product type, and product detail.

The dataset provides sufficient information to analyze sales performance across different stores, products, categories, and hours of the day.

Table 4.1: Dataset Attributes

Attribute	Description
Transaction ID	Unique identifier of the transaction
Year	Year in which the transaction occurred
Transaction Time	Time at which the transaction occurred
Transaction Quantity	Number of items purchased
Store ID	Unique identifier of the store
Store Location	Location of the coffee store
Product ID	Unique product identifier
Unit Price	Price of one unit of the product
Product Category	Category of the product
Product Type	Type of coffee or beverage
Product Detail	Specific product name/detail

4.3 Data Preprocessing

Data preprocessing was performed to ensure that the dataset was suitable for analysis. The transaction time field was converted into an appropriate time format to enable time-based analysis.

Additional analytical variables were created from the existing transaction information. In particular, the transaction hour was extracted from the transaction time field. This enabled the identification of hourly sales and revenue patterns.

Revenue was calculated using transaction quantity and unit price. The calculated revenue field was subsequently used for store-level, category-level, product-level, and hourly revenue analysis.

The processed dataset was then stored in CSV format and used as the input for the analytical dashboard.

4.4 Data Analysis

The processed dataset was analyzed using descriptive and exploratory data analysis techniques. The analysis focused on identifying important business performance indicators and understanding sales patterns.

The major analytical areas included total revenue, total transactions, total quantity sold, average transaction value, store performance, product category performance, individual product performance, and hourly revenue.

Grouping and aggregation techniques were used to compare revenue across stores, categories, products, and hours. These analyses helped identify the highest-performing business segments and periods of peak demand.

4.5 Data Visualization

Data visualization was used to communicate the analytical findings in an easily understandable form. Bar charts and other graphical representations were created to compare revenue and sales performance across different dimensions.

The major visualizations include revenue by store location, revenue by product category, top-performing products, and hourly revenue. These visualizations allow users to quickly identify important trends and differences in business performance.

4.6 Dashboard Development

An interactive dashboard was developed using Streamlit to present the results of the analysis. The dashboard integrates the processed dataset, key performance indicators, analytical results, and visualizations into a single interface.

The dashboard displays important metrics such as total revenue, total transactions, total quantity sold, average transaction value, best-performing store, best-performing category, best-performing product, and peak revenue hour.

The dashboard is designed to provide a simple and accessible interface through which users can explore coffee sales performance and demand patterns.

4. METHODOLOGY

STEP 4.1- Project Workflow

The project follows a structured data analytics workflow to transform raw coffee transaction data into meaningful business insights. The methodology begins with understanding and loading the transaction dataset, followed by data inspection and preprocessing. The cleaned data is then used for feature engineering and exploratory analysis.

The analysis focuses on transaction volume, quantity sold, revenue, product performance, store performance, product categories, and hourly demand patterns. Key performance indicators are calculated to provide a summarized view of business performance.

Finally, the analytical results are presented through visualizations and an interactive Streamlit dashboard. The overall workflow ensures that the raw transaction data is systematically converted into actionable information for sales and demand analysis.

Project workflow:

Raw Dataset → Data Cleaning → Feature Engineering → Exploratory Data Analysis →
Business Analysis → Visualization → Interactive Dashboard → Insights

STEP 4.2 — Data Collection

The dataset used for this project is the transaction-level sales dataset of Afficionado Coffee Roasters. The dataset contains individual coffee shop transactions along with information related to transaction time, quantity, store location, product details, product category, and unit price.

The dataset consists of 149,116 transaction records. These records provide sufficient information to analyze sales performance across different stores, products, categories, and hours of operation.

The dataset was provided in spreadsheet format and was processed using Python and Pandas for further analysis.

Attribute	Description
Dataset	Afficionado Coffee Roasters
Number of Transactions	149,116
Transaction Quantity	Number of products purchased
Store Location	Location of the coffee store
Product Category	Category of the product
Product Type	Type of coffee/product
Product Detail	Specific product purchased
Unit Price	Price per unit
Transaction Time	Time of transaction

STEP 4.3 — Data Preprocessing

Data preprocessing was performed to ensure that the dataset was suitable for analysis. The raw transaction data was first loaded into a Pandas DataFrame and its structure, data types, and columns were examined.

The transaction time field was converted into an appropriate time format to enable time-based analysis. Relevant numerical and categorical fields were examined for consistency. Additional checks were performed to ensure that the dataset could be reliably used for revenue, product, store, and hourly analysis.

The preprocessing stage resulted in a structured dataset that could be used for subsequent analytical operations and visualization.

Major preprocessing operations

1. Loading the transaction dataset.
2. Inspecting dataset dimensions and column names.
3. Checking data types.
4. Converting transaction time into a usable time format.
5. Identifying relevant numerical and categorical variables.

6. Preparing the dataset for feature engineering.
7. Saving the processed dataset for further analysis.

STEP 4.4 — Feature Engineering

Feature engineering was performed to derive additional variables required for business analysis. A major feature created during this stage was the transaction hour, which was extracted from the transaction time field.

A revenue variable was also calculated using transaction quantity and unit price. This allowed the project to analyze not only the number of products sold but also the monetary contribution generated by each transaction.

Revenue Calculation

Revenue = Transaction Quantity × Unit Price

Hour Extraction

Hour = Hour component of Transaction Time

For example:

Transaction Time	Hour	Quantity	Unit Price	Revenue
07:06:11	7	2	3.00	6.00
07:08:56	7	2	3.10	6.20
07:14:04	7	2	4.50	9.00

STEP 4.5 — Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the underlying patterns in the coffee sales dataset. The analysis examined overall revenue, transaction volume, quantity sold, store performance, product category performance, individual product performance, and hourly demand.

Aggregation techniques were used to summarize the transaction-level data into meaningful business-level metrics. Grouping operations were performed across stores, categories, products, and transaction hours.

Visualizations were then created to identify trends and differences in sales performance. These visualizations formed the analytical foundation for the final interactive dashboard.

Total Revenue Analysis

- Total Transaction Analysis
- Quantity Sold Analysis
- Store Performance Analysis
- Product Category Analysis
- Product Performance Analysis
- Hourly Revenue Analysis
- Hourly Demand Analysis

STEP 4.6 — Business Analysis

The processed dataset was analyzed from a business perspective to identify important sales and demand patterns. Key Performance Indicators (KPIs) were calculated to provide a concise summary of overall business performance.

The analysis identified the highest-performing store, highest-performing product category, highest-revenue product, and peak revenue hour. These findings provide useful information for understanding customer purchasing behavior and identifying periods of high business activity.

KPI	Result
Total Revenue	698,812.33
Total Transactions	149,116
Total Quantity Sold	214,470
Average Transaction Value	4.69
Best Store	Hell's Kitchen
Best Category	Coffee
Best Product	Sustainably Grown Organic Lg

STEP 4.7 — Dashboard Development

An interactive dashboard was developed using Streamlit to present the analytical findings in a user-friendly format. The dashboard integrates key performance indicators, transaction-level data, revenue analysis, store performance, product category analysis, and hourly demand analysis.

The dashboard allows users to interact with the dataset and examine business performance through tables, filters, and visualizations. The objective of the dashboard is to make the analytical results easier to understand and support data-driven decision-making.

5. DATA ANALYSIS AND RESULTS

5.1 Key Performance Indicators

Key Performance Indicators (KPIs) were calculated to provide a summarized view of the overall performance of Afficionado Coffee Roasters. These indicators help evaluate transaction activity, sales volume, revenue generation, and average transaction performance.

KPI	Value
Total Revenue	₹698,812.33
Total Transactions	149,116
Total Quantity Sold	214,470
Average Transaction Value	₹4.69
Best Performing Store	Hell's Kitchen
Best Product Category	Coffee
Best Performing Product	Sustainably Grown Organic Lg
Peak Revenue Hour	10:00 AM

5.2 — Revenue Analysis

Revenue analysis was conducted to understand the overall monetary performance of the business. Revenue was calculated at the transaction level using transaction quantity and unit price. The aggregated transaction revenue was then used to determine total revenue and compare revenue performance across stores, product categories, products, and hours of operation.

The total revenue generated from the analyzed transactions was ₹698,812.33. This provides an overall measure of the sales generated during the period covered by the dataset.

Product Category Distribution

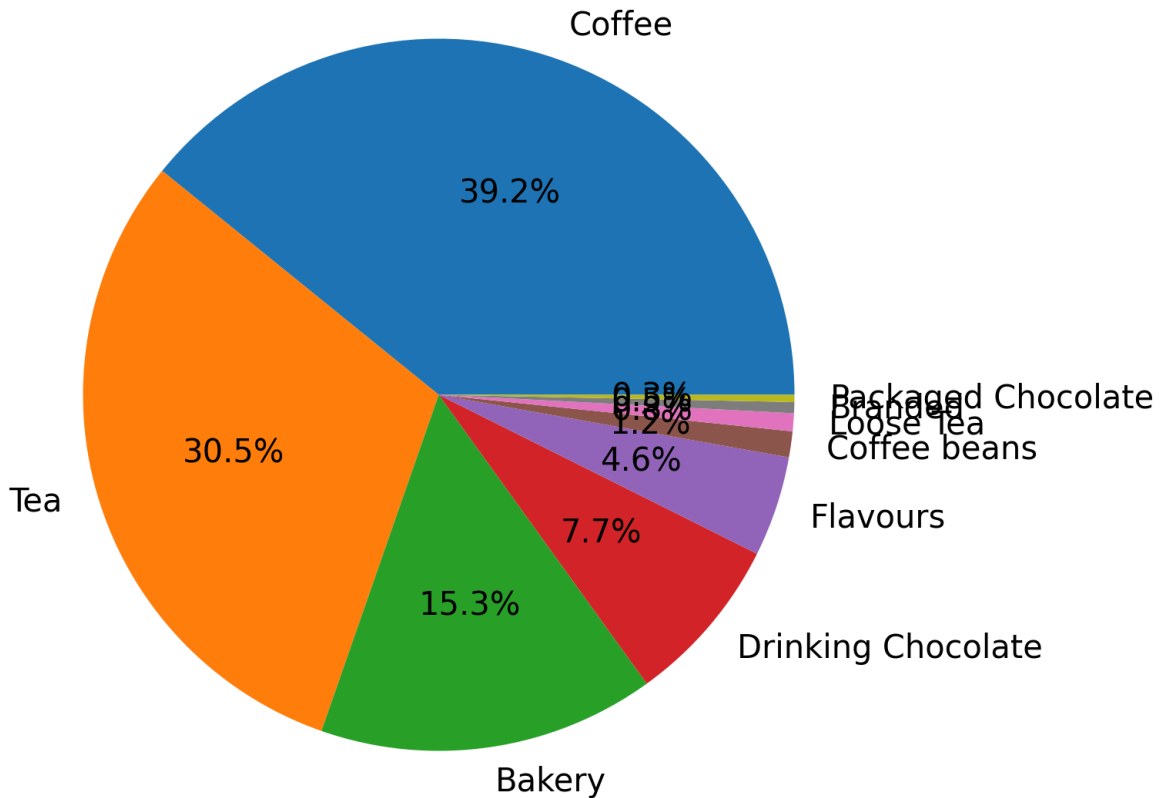


Figure 5.1: Revenue by Product Category

The category-level revenue analysis shows the relative contribution of different product categories to overall sales. The analysis indicates that Coffee is the highest-performing product category in terms of revenue.

5.3 Store Performance Analysis

Store-level analysis was performed to identify differences in revenue generation across the coffee shop locations. Revenue was aggregated based on store location to determine the contribution of each store to overall business performance.

The analysis identified Hell's Kitchen as the best-performing store based on revenue generation. This indicates that the location generated the highest revenue among the stores included in the dataset.

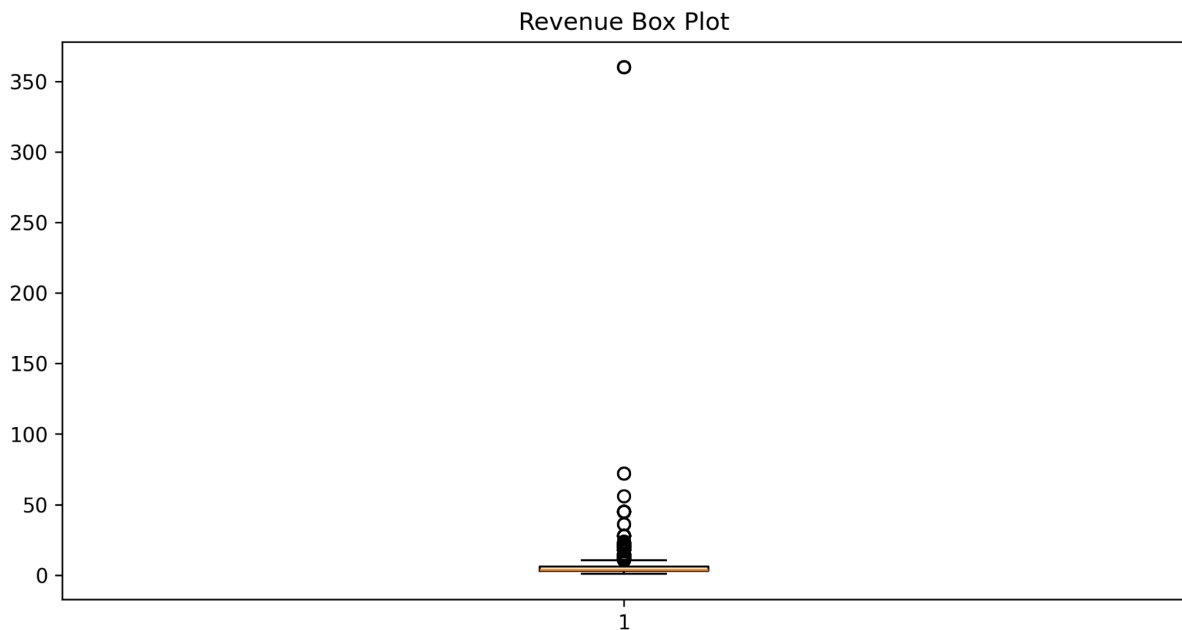


Figure 5.2: Revenue by Store Location

The visualization provides a comparison of revenue generated by individual store locations and helps identify the strongest and weaker-performing locations.

5.4 Product Performance Analysis

Product-level analysis was conducted to identify the products contributing most significantly to overall revenue. Revenue was aggregated for individual products and ranked to identify the highest-performing products.

The analysis identified Sustainably Grown Organic Lg as the highest-revenue product in the analyzed dataset.

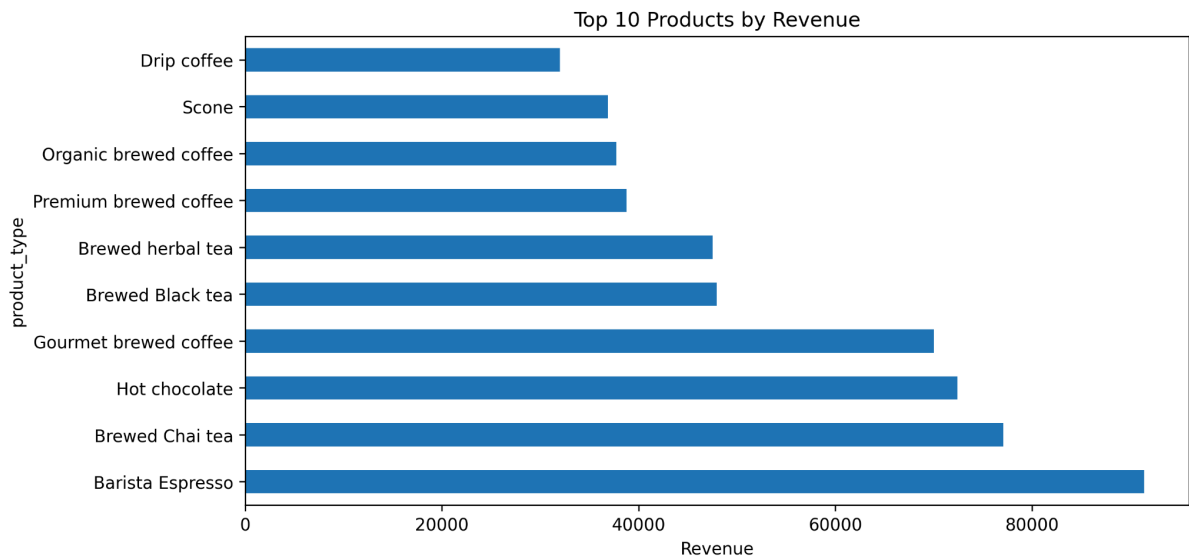


Figure 5.3: Top 10 Products by Revenue

The visualization presents the top-performing products based on revenue. Such analysis can help management understand which products contribute significantly to sales and can support decisions related to product promotion, inventory planning, and product availability.

5.5 Hourly Revenue Analysis

Hourly revenue analysis was conducted to understand how sales activity changes throughout the operating day. The transaction time was converted into an hourly variable, allowing total revenue to be aggregated for each hour.

The analysis identified 10:00 AM as the peak revenue hour. This indicates that the business experiences particularly strong revenue activity during the morning period.

Understanding hourly revenue patterns can help businesses optimize staffing, inventory availability, promotional activities, and operational planning during high-demand periods.

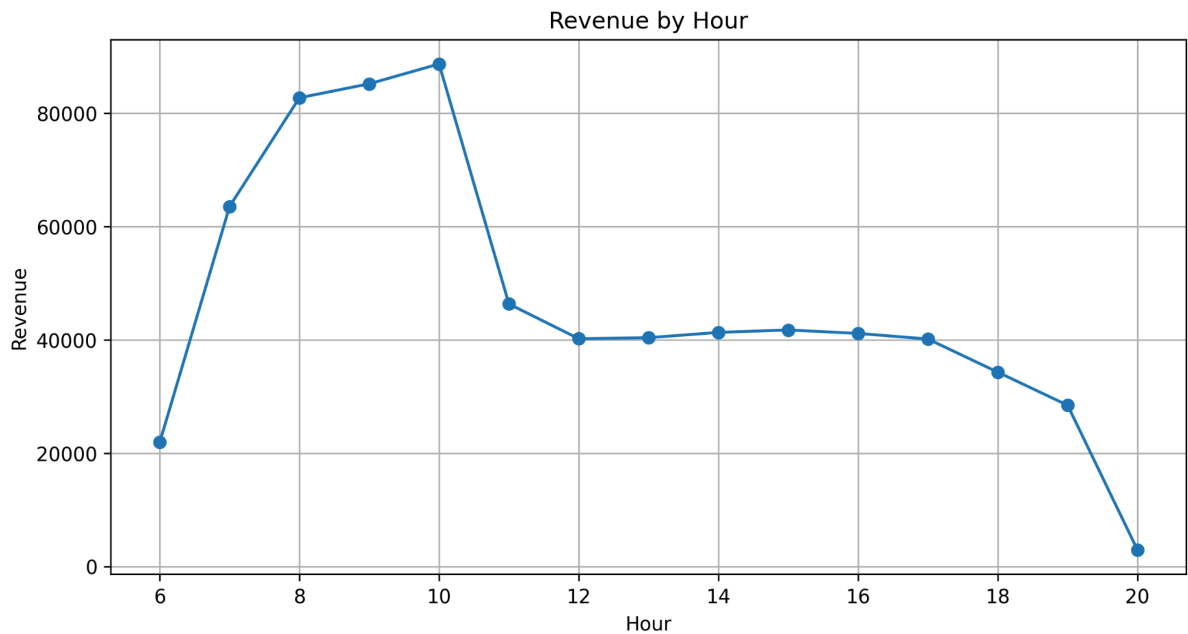


Figure 5.4: Hourly Revenue Analysis

5.6 Transaction and Quantity Analysis

Transaction and quantity analysis was performed to understand the overall purchasing activity recorded in the dataset. The analysis considered both the total number of transactions and the total quantity of products sold.

A total of 149,116 transactions were recorded in the dataset, with a total quantity sold of 214,470 units. This indicates that customers purchased multiple products across a significant number of transactions.

The comparison between transaction count and quantity sold provides an understanding of customer purchasing volume. These metrics can help the business evaluate sales activity and understand the overall scale of product movement during the analyzed period.

Metric	Result
Total Transactions	149,116
Total Quantity Sold	214,470
Average Quantity per Transaction	1.44

Table 5.1: Transaction and Quantity Summary

5.7 Interactive Dashboard Analysis

An interactive dashboard was developed to present the results of the sales and demand analysis in a simple and accessible format. The dashboard combines key performance indicators, transaction data, revenue analysis, store performance, product analysis, and hourly revenue patterns.

The dashboard provides users with an interactive interface for exploring the dataset and understanding important business metrics. It allows analytical results to be presented visually rather than relying only on numerical tables.

The dashboard was developed using Streamlit, while Pandas was used for data processing and analysis. The visualizations provide a summarized view of business performance and help users identify important trends and patterns.

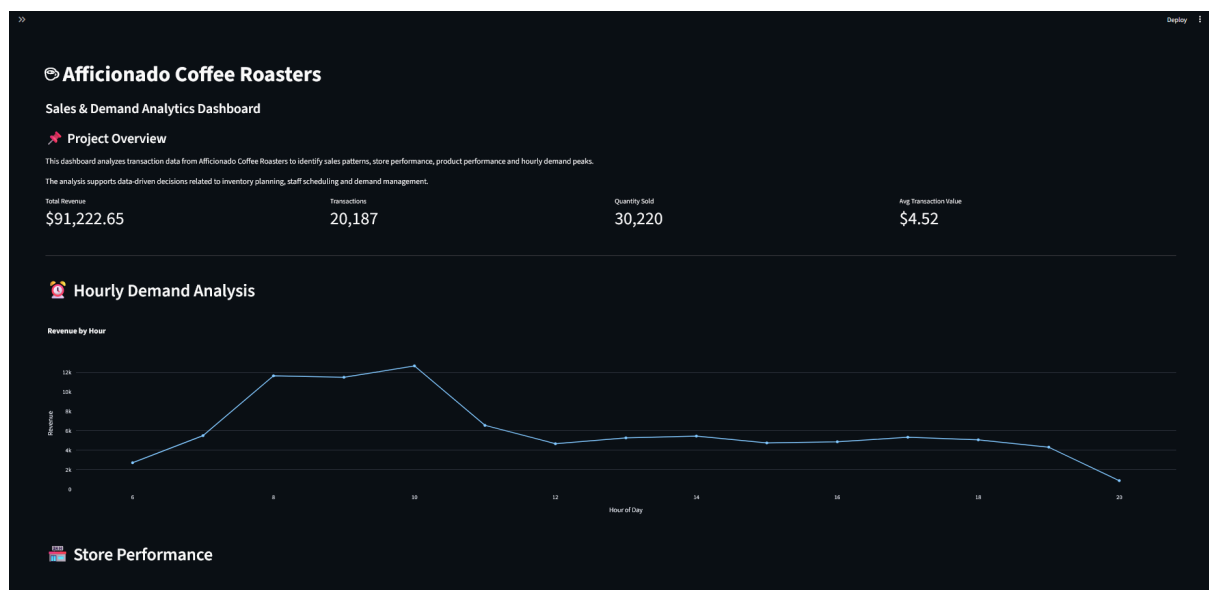


Figure 5.5: Interactive Sales and Demand Analytics Dashboard

The dashboard provides a consolidated view of the major analytical findings of the project. The KPI section summarizes overall business performance, while the visualizations provide detailed information about revenue, stores, products, categories, and hourly demand.

5.8 Key Findings

The analysis of the Afficionado Coffee Roasters transaction dataset produced several important findings. The major findings obtained from the analysis are summarized below:

1. The dataset contains 149,116 transactions and records a total quantity sold of 214,470 units.
2. The total revenue generated from the analyzed transactions was ₹698,812.33.
3. The average transaction value was approximately ₹4.69.
4. Hell's Kitchen was identified as the highest-performing store based on revenue.
5. Coffee was identified as the highest-performing product category.
6. Sustainably Grown Organic Lg was identified as the highest-revenue individual product.
7. 10:00 AM was identified as the peak revenue hour.
8. Hourly analysis demonstrates that sales activity varies considerably throughout the operating day.
9. The interactive dashboard provides a convenient way to explore these findings and supports easier interpretation of the underlying transaction data.

5.9 Business Insights

The results of the analysis provide several insights that can support business decision-making. The identification of the highest-performing store can help management understand which locations are generating stronger sales and investigate the factors contributing to their performance.

The strong performance of the Coffee category indicates that coffee products represent an important contributor to overall revenue. Similarly, identifying the highest-revenue product can help management prioritize product availability and consider targeted promotional strategies.

The identification of 10:00 AM as the peak revenue hour suggests that operational resources should be appropriately planned during periods of high customer activity. Staffing levels, inventory availability, and service capacity can be adjusted according to observed demand patterns.

Overall, the analysis demonstrates how transaction-level data can be transformed into meaningful business insights through data cleaning, feature engineering, aggregation, visualization, and dashboard development.

6. CONCLUSION AND RECOMMENDATIONS

6.1 Conclusion

This project successfully analyzed transaction-level sales data from Afficionado Coffee Roasters to identify important patterns in revenue, transaction activity, product performance, store performance, and hourly demand.

The analysis was performed using Python and Pandas, with data preprocessing and feature engineering techniques used to prepare the dataset for analysis. Revenue was calculated from transaction quantity and unit price, while transaction time was used to analyze hourly sales patterns.

The analysis identified a total revenue of ₹698,812.33 from 149,116 transactions, with a total quantity sold of 214,470 units. The average transaction value was approximately ₹4.69. Hell's Kitchen was identified as the highest-performing store, while Coffee was the leading product category. Sustainably Grown Organic Lg was identified as the highest-revenue product, and 10:00 AM was identified as the peak revenue hour.

An interactive Streamlit dashboard was also developed to present the analytical results in a user-friendly format. The dashboard brings together key performance indicators and visualizations, making the results easier to interpret and explore.

Overall, the project demonstrates how data analytics can transform raw transaction records into actionable business insights. These insights can support better operational planning, inventory management, staffing decisions, and product-level decision-making.

6.2 Recommendations

Based on the findings obtained from the analysis, the following recommendations can be considered by Afficionado Coffee Roasters:

1. Optimize Staffing During Peak Hours

Since 10:00 AM was identified as the peak revenue hour, management can consider allocating additional staff and operational resources during high-demand periods to maintain service efficiency.

2. Maintain Availability of High-Performing Products

Products generating higher revenue should receive appropriate inventory attention to reduce the possibility of stock-outs during periods of high demand.

3. Focus on High-Performing Categories

The strong performance of the Coffee category suggests that management can continue prioritizing popular coffee products through availability, promotions, and product placement.

4. Analyze Store-Level Differences

Since Hell's Kitchen was identified as the highest-performing store, management can investigate the factors contributing to its performance and determine whether successful practices can be applied to other locations.

5. Use Data for Operational Planning

Historical sales and hourly demand patterns can be used to support staffing and inventory planning instead of relying entirely on intuition or uniform resource allocation.

6.3 Project Limitations

The project has certain limitations that should be considered when interpreting the results. The available dataset contains transaction-level information but does not provide a complete date field, which limits the ability to perform detailed daily, weekly, monthly, and long-term time-series analysis.

Additionally, the dataset does not include information about factors such as promotions, weather conditions, holidays, special events, customer demographics, or inventory availability. These factors may influence customer demand but could not be incorporated into the current analysis.

Therefore, the findings primarily describe historical transaction and sales patterns available within the provided dataset.

6.4 Future Scope

The current project can be extended into a more advanced demand forecasting and predictive analytics system. Future development could incorporate a complete transaction date field to enable daily, weekly, and monthly time-series forecasting.

Advanced forecasting models such as ARIMA, SARIMA, Exponential Smoothing, Prophet, or Gradient Boosting could be evaluated to predict future sales and demand.

Additional external variables such as holidays, weather, promotions, special events, and store operating conditions could also be incorporated to improve forecasting accuracy.

Future versions of the Streamlit dashboard could include forecast horizons, prediction intervals, store-level demand forecasting, hourly demand heatmaps, model comparison, and best-case and worst-case demand scenarios.

These improvements could transform the current descriptive analytics solution into a comprehensive predictive demand-planning system.

7. CONCLUSION

7.1 Conclusion

The Aficionado Coffee Roasters Sales and Demand Analytics project successfully demonstrates how data analytics can be applied to a real-world coffee sales dataset to identify important business patterns and generate meaningful insights.

The project involved data preparation, data cleaning, exploratory analysis, feature engineering, revenue analysis, product analysis, store performance analysis, and hourly demand analysis. The processed transaction data was transformed into meaningful analytical metrics and visualizations to support business decision-making.

The analysis identified important performance indicators including total revenue, total transactions, total quantity sold, average transaction value, best-performing store, leading product category, highest-revenue product, and peak revenue hour. These findings provide a clear understanding of the historical sales performance of the business.

An interactive Streamlit dashboard was also developed to present the analytical results in a simple and user-friendly format. The dashboard enables users to explore transaction data, revenue performance, product categories, store performance, and hourly demand patterns without requiring technical knowledge of Python or data analysis.

Overall, the project demonstrates the practical value of data analytics in converting large transaction datasets into actionable business information. The developed solution can help management understand sales patterns, identify high-performing areas, and make more informed operational decisions.

7.2 Key Findings

The major findings obtained from the analysis are summarized below:

1. Total Revenue: The dataset generated a total revenue of approximately 698,812.33.
2. Total Transactions: A total of 149,116 transactions were recorded.
3. Total Quantity Sold: The analysis recorded approximately 214,470 units sold.

4. Average Transaction Value: The average transaction value was approximately 4.69.
5. Best Performing Store: Hell's Kitchen generated the highest revenue among the analyzed store locations.
6. Best Product Category: Coffee was the highest-revenue product category.
7. Best Performing Product: Sustainably Grown Organic Lg generated the highest revenue among individual products.
8. Peak Revenue Hour: 10:00 AM was identified as the peak revenue-generating hour.

These findings demonstrate clear differences in revenue contribution across stores, product categories, individual products, and hours of operation.

7.3 Recommendations

Based on the analytical findings, the following recommendations can be considered:

1. Optimize staffing during peak hours

Since 10:00 AM represents the highest revenue-generating hour, adequate staffing should be maintained during peak demand periods to reduce waiting time and improve customer service.

2. Focus on high-performing products

Products generating higher revenue should receive greater attention through inventory planning, product placement, and promotional activities.

3. Analyze store-level performance

High-performing stores can be studied to identify successful operational practices that may be applied to other locations.

4. Improve inventory planning

Hourly demand patterns can help management anticipate periods of higher sales and maintain appropriate inventory levels.

5. Use the dashboard for regular monitoring

The interactive dashboard can be used as a management-support tool for continuously monitoring important sales and revenue indicators.

6. Develop predictive forecasting in the future

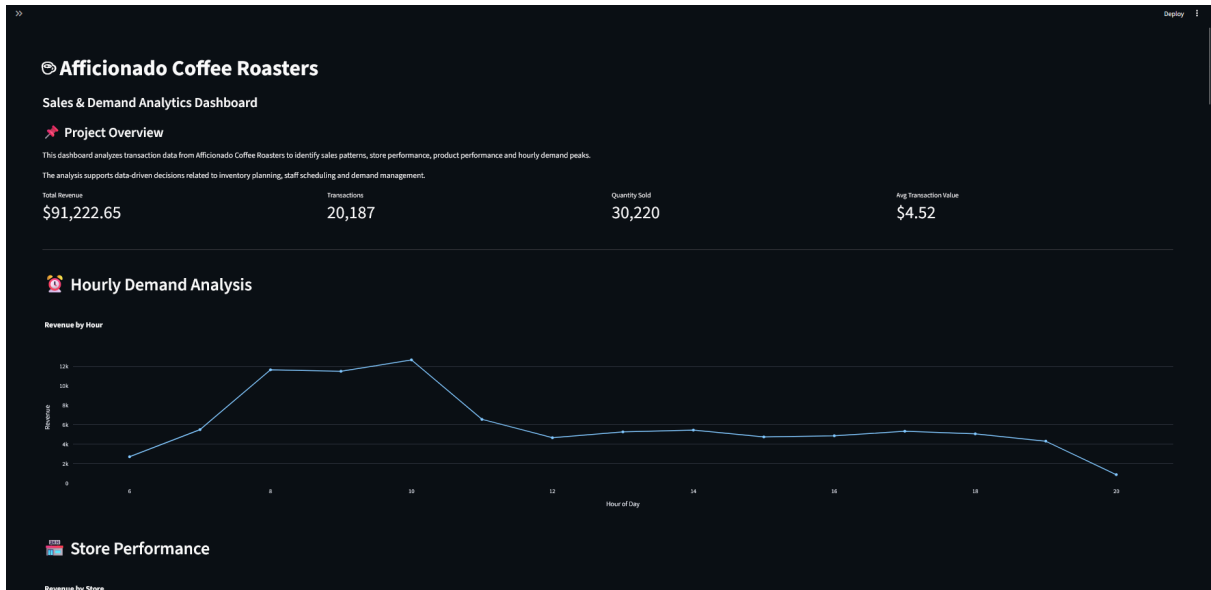
Future versions of the project can incorporate complete date information and advanced forecasting techniques to predict future sales and customer demand.

8. REFERENCES

1. Python Documentation — Python programming language and standard libraries.
2. Pandas Documentation — Data manipulation and analysis.
3. Matplotlib Documentation — Data visualization.
4. Streamlit Documentation — Interactive dashboard development.
5. NumPy Documentation — Numerical and scientific computing.
6. Dataset: Aficionado Coffee Roasters Transactions Dataset.

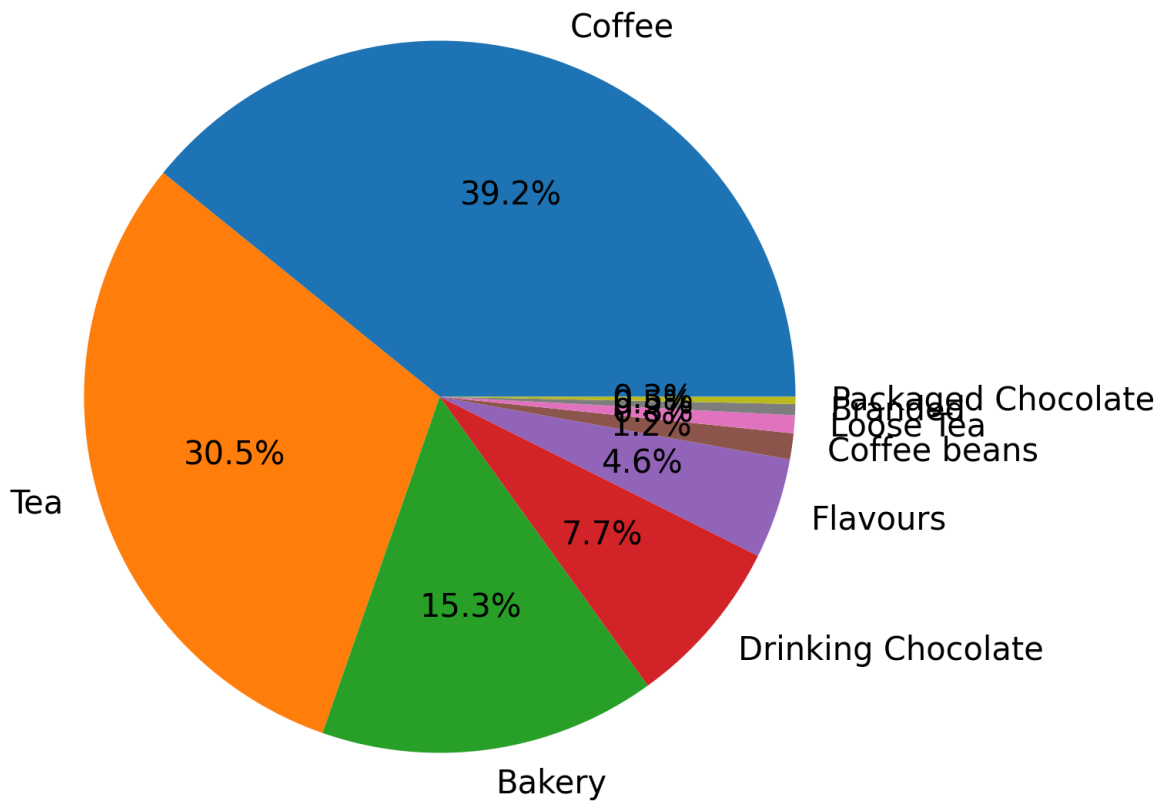
APPENDIX A — PROJECT OUTPUTS

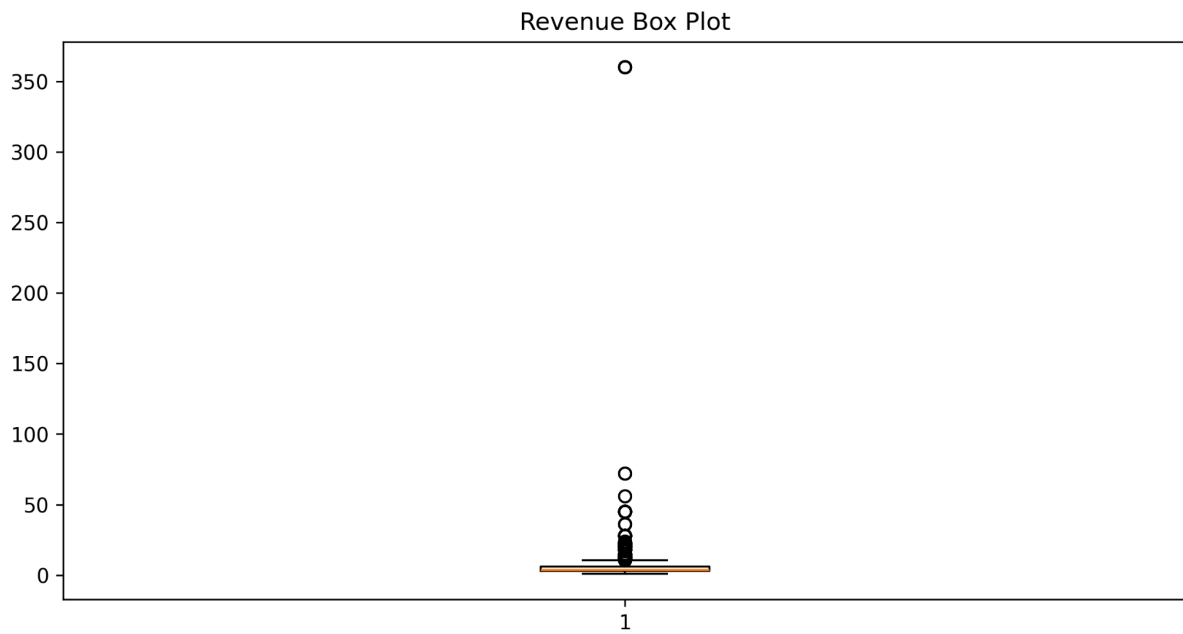
A.1 Streamlit Dashboard



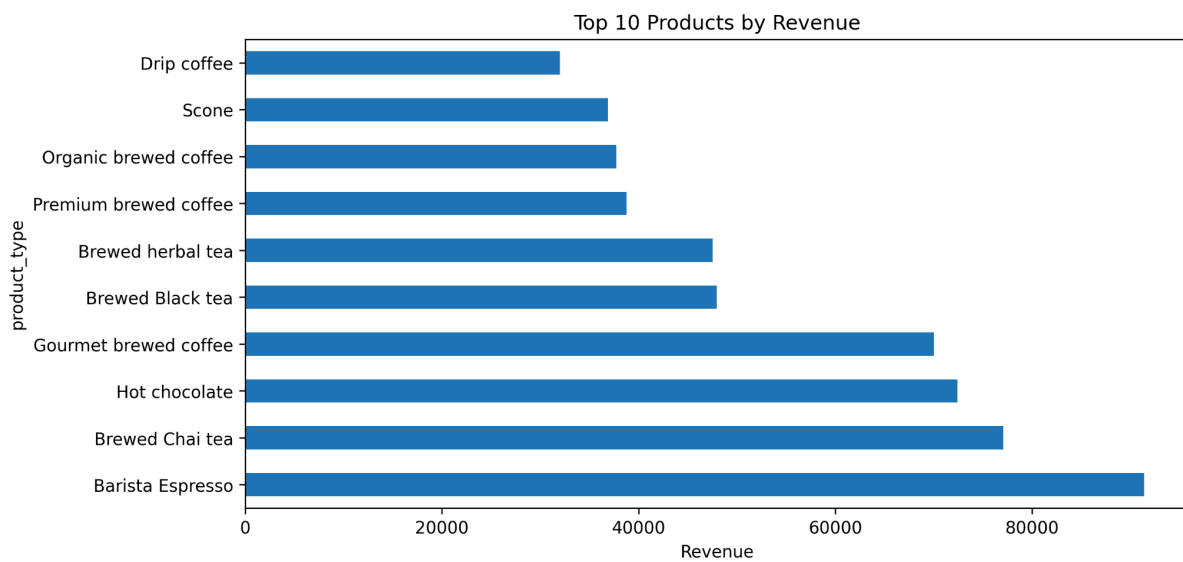
A.2 Revenue Analysis

Product Category Distribution

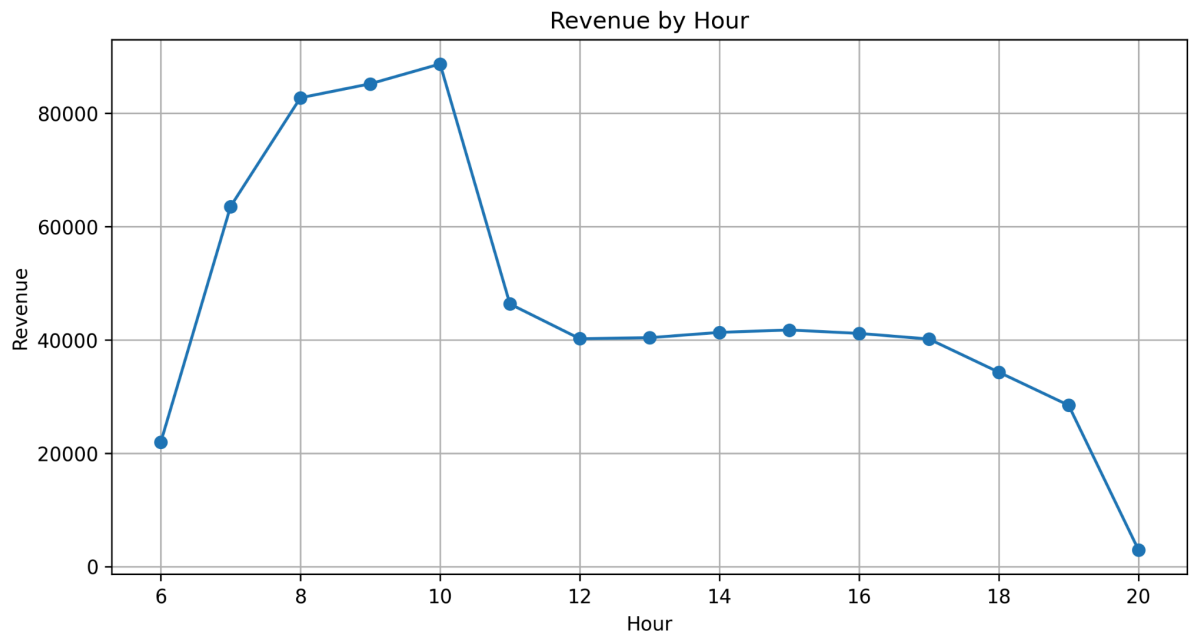




A.3 Product Analysis



A.4 Hourly Revenue Analysis



A.5 Final Project Results

Files

..

config

drive

images

sample_data

Afficionado Coffee Roasters.xlsx ...

Coffee_Feature_Engineered.csv

Coffee_Final_Analysis.csv

Coffee_Hourly_Analysis.csv

Coffee_cleaned.csv

app.py

logs.txt

revenue_by_category.png

revenue_by_store.png

revenue_top10_products.png

```
"Best Store:",
df.groupby("store_location")["Revenue"]
.sum()
.idxmax()

)

print(
"Best Category:",
df.groupby("product_category")["Revenue"]
.sum()
.idxmax()

)

print(
"Best Product:",
df.groupby("product_detail")["Revenue"]
.sum()
.idxmax()

)

print(
"Peak Revenue Hour:",
df.groupby("hour")["Revenue"]
.sum()
.idxmax()

)
```

*** ----- FINAL PROJECT RESULTS ----- ***
Total Revenue: 698812.33
Total Transactions: 149116
Total Quantity Sold: 214470
Average Transaction Value: 4.69
Best Store: Hell's Kitchen
Best Category: Coffee
Best Product: Sustainably Grown Organic Ig
Peak Revenue Hour: 10